Undeciphered Ancient Scripts and Languages

Reference document for Glossa Lab research planning.

This document catalogues all known undeciphered or partially deciphered ancient scripts and proto-writing systems, organised by region. It is maintained to:

- Drive experiment prioritisation (which scripts could benefit from computational analysis)
- Identify corpus acquisition targets
- Track the decipherment state of each script so we do not duplicate published work
- Note which scripts are plausible future Tier 5 hypothesis-test candidates

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European Scripts

Script	Location	Date	Status	Corpus availability
Cretan Hieroglyphic	Crete	c. 2100 BCE	Completely undeciphered	~300 inscribed objects; CHIC corpus published
Linear A	Aegean (Crete)	c. 1800–1450 BCE	Script signs known via Linear B correspondences; underlying language unknown	~1,500 inscriptions; SigLA database
Phaistos Disc	Crete	c. 1850–1300 BCE	Completely undeciphered; no consensus on any reading	Single object, 242 signs; photographed
Cypro-Minoan	Cyprus	c. 1550 BCE	Undeciphered	~230 inscribed objects; DĀMOS partial
Grakliani Hill Script	Georgia	c. 11th–10th century BCE	Undeciphered	Single inscription
SW Paleohispanic Script	Iberian Peninsula	c. 700 BCE	Partially deciphered; sign values proposed, language unknown	~100 inscriptions; published corpora
Paleohispanic Scripts	Iberian Peninsula	c. 700 BCE onward	Several variants; some partially read	Varied; MLH / HESPERIA database
Sitovo Inscription	Bulgaria	c. 300–100 BCE	Undeciphered	Single inscription
Alekanovo Inscription	Russia	c. 10th–11th century CE	Undeciphered	Single object
Rohonc Codex	Hungary	17th–19th centuries	Completely undeciphered; authorship and language unknown	~450 pages; scanned

Script	Location	Date	Status	Corpus availability
Voynich Manuscript	Medieval Europe	c. 1404–1438 (radiocarbon)	Completely undeciphered; possibly a hoax or cipher	~240 pages; fully digitised at Beinecke Library
Pisa Baptistery Inscription	Italy	Medieval	Undeciphered religious text fragment	Single inscription

Near Eastern / Asian Scripts

Script	Location	Date	Status	Corpus availability
Proto-Elamite	Iran (Susa)	c. 3200–2900 BCE	Completely undeciphered; one of the oldest writing systems	~5,000 tablets; CDLI database
Linear Elamite	Iran	c. 2200–1850 BCE	Partially deciphered (2022 Desset et al. partial reading)	~40 inscriptions
Indus / Harappan Script	Pakistan / India	c. 3500–2000 BCE	Completely undeciphered; possibly non-linguistic or Dravidian	~4,000+ inscriptions; ICIT / Mahadevan M77
Khitan Large Script	Manchuria / NE Asia	10th–12th centuries CE	Largely undeciphered	Several hundred texts
Khitan Small Script	Manchuria	10th–12th centuries CE	Mostly undeciphered; some morphological analysis done	Several hundred texts
Issyk Inscription	Kyrgyzstan	c. 5th–3rd century BCE	Undeciphered; Saka-Scythian context	Single gold vessel inscription
Ba–Shu Scripts	Ancient China (Sichuan)	Various	Undeciphered	Limited bronze inscriptions
Tujia Script	China	Various (folk script)	Undeciphered or unanalysed	Some museum holdings

Mesoamerican Scripts

Script	Location	Date	Status	Corpus availability
Isthmian / Epi-Olmec	Mexico (southern coast)	c. 900–200 BCE	Largely undeciphered; some proposed readings	~7 known inscriptions; Stela C, La Mojarra

Script	Location	Date	Status	Corpus availability
Zapotec Writing	Oaxaca, Mexico	c. 600 BCE onward	Partially deciphered; calendar/day-signs readable, rest unclear	Several hundred inscriptions
Mixtec Writing	Oaxaca / Puebla, Mexico	c. 900–1600 CE	Partially deciphered; many codices and pictographic elements read	Several codices; pictographic rather than phonetic
Cascajal Block	Veracruz, Mexico	c. 900–800 BCE	Undeciphered; authenticity accepted but no consensus reading	Single object, 62 signs

African Scripts

Script	Location	Date	Status	Corpus availability
Meroitic Script	Sudan (Kingdom of Kush)	c. 300 BCE – 350 CE	Sign values known; underlying language poorly understood	~1,200 inscriptions; REM database
Wadi el-Hol Inscriptions	Egypt (Western desert)	c. 1900–1800 BCE	Early proto-Sinaitic / early alphabetic; proposed readings, no consensus	2 main inscriptions
Nsibidi	Nigeria / Cameroon	Origin uncertain	Ideographic; not phonetic writing; indigenous use continues	Some collected by ethnographers
Proto-Saharan / Tifinar precursors	Sahara	Neolithic onward	Disputed as true writing; related to modern Tifinagh?	Rock art corpora

Oceanic Scripts

Script	Location	Date	Status	Corpus availability
Rongorongo (Kōhau Rongorongo)	Easter Island (Rapa Nui)	Attested by 19th century	Completely undeciphered; oral tradition lost after 1862 slave raids	~25 surviving objects; photographed; Fischer 1997 corpus
Kohi Script	Polynesia (claimed)	Various	Disputed authenticity; largely unknown	Very limited

Neolithic / Proto-Writing Systems *(Disputed as true writing)*

These are disputed as constituting true writing rather than decorative or counting symbols.

Symbol System	Location	Date	Status
Vinča / Old European Script	Southeast Europe (near Belgrade)	c. 6000–4500 BCE	Highly disputed; many researchers consider these non-linguistic marks
Jiahu Symbols	China (Henan)	c. 8600–7600 BCE	Questioned whether truly writing or decorative; possibly proto-numerals
Banpo Symbols	China (Xi'an region)	c. 4500–3750 BCE	Possibly proto-writing; no linguistic interpretation accepted
Dispilio Tablet	Greece	c. 5260 BCE	Early symbols; tablet submerged and partially damaged; status debated
Megalithic Graffiti	Various European sites	Neolithic	Cup marks and carved symbols; status as writing not established

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Three Categories of Decipherment Difficulty

Scholars recognise three primary challenge types, which drive different experimental strategies:

1 — Unknown Script, Known Language

The underlying language is related to a known language but the sign system is unfamiliar.

- **Examples:** Rongorongo (likely Polynesian), Zapotec (related to modern Zapotec languages), Cypro-Minoan (likely related to Mycenaean Greek)
- **Strategy:** Use modern/attested relative as language model; apply phoneme-substitution / bigram matching. This is the Ugaritic→Hebrew pattern (Tier 1a/1b) and is where our beam engine has the highest expected yield.

2 — Known Script, Unknown Language

The script has been deciphered (sign values known), but the language it encodes remains obscure.

- **Examples:** Etruscan (uses Greek alphabet, language unrelated to known families), Meroitic (sign values known since 1909, language only partially understood)
- **Strategy:** Use known phoneme inventory to test language-family hypotheses via Z-score / KL-divergence method. This is the Tier 5 pattern.

3 — Unknown Script, Unknown Language

Both the sign system and the underlying language are completely opaque.

- **Examples:** Indus script, Linear A, Proto-Elamite, Cretan Hieroglyphic, Phaistos Disc
- **Strategy:** Structural analysis (entropy, sign classification, Ventris affinity) + multi-hypothesis testing with many candidate language families simultaneously. Most difficult category; Tiers 3–5 of our programme apply here.

Why These Remain Undeciphered

The fundamental obstacle is the **absence of a Rosetta Stone equivalent** — a bilingual or multilingual text that allows symbols to be anchored to a known language. Additional barriers:

1. **Small corpus** — Limited surviving texts make statistical pattern recognition unreliable. The Phaistos Disc (242 signs total) is essentially impossible to decipher computationally.
2. **Unknown language** — The underlying language may be extinct with no surviving relatives (e.g. Minoan, Proto-Elamite).
3. **Loss of context** — Slave raids (Rongorongo), deliberate destruction (Maya codices, partially), and site destruction remove crucial usage context.
4. **Disputed authenticity** — Vinča symbols, Jiahu marks, and the Cascajal Block have contested status as writing.
5. **No anchorpoints** — No identifiable proper names, numerals, calendrical dates, or other recognisable elements to bootstrap readings.
6. **Logo-syllabic complexity** — Scripts mixing logograms with phonetic signs (Indus, Proto-Elamite, Isthmian) resist pure phonological analysis.

Glossa Lab Research Priority Map

Based on the criteria: *corpus size ≥ 200 inscriptions · known or plausible language family · digitised corpus available*

Priority	Script	Rationale
Active (Tier 5)	Indus / Harappan	Large corpus (4,000+), Dravidian hypothesis strong, ICIT data held
Active (Tier 4)	Linear A	SigLA corpus; Linear B correspondences give partial anchor
Next target	Meroitic	Script readable; ~1,200 inscriptions; Nilo-Saharan / Cushitic LMs available
Next target	Linear Elamite	Partial decipherment (2022) gives anchor points; ~40 inscriptions
Future	Cretan Hieroglyphic	~300 objects; language unknown but Minoan = likely substrate
Future	Rongorongo	~25 objects; corpus tiny but Polynesian LM excellent
Future	Proto-Elamite	5,000 tablets; completely undeciphered; numeric/administrative context only
Future	Cypro-Minoan	230 objects; possible Greek link gives phonological anchor
Research only	Voynich, Phaistos Disc	Corpus too small or authenticity/language too uncertain for statistical approach

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Data Sources and Corpora

Script	Primary Source	URL / Notes

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Linear A	SigLA (Signary of Linear A)	sigla.classics.ox.ac.uk
Linear A raw	tylerlengyel.com phase1 data	Used in current experiments
Indus	ICIT (Fuls 2023)	Physical corpus; Dr. Fuls collaboration
Indus	Mahadevan M77 concordance	Mahadevan (1977); OCR'd in experiments
Proto-Elamite	CDLI (Cuneiform Digital Library Initiative)	cdli.ucla.edu
Meroitic	REM (Répertoire d'épigraphie méroïtique)	Published; digital version in progress
Rongorongo	Fischer (1997) corpus	25 objects catalogued
Cretan Hieroglyphic	CHIC (Corpus Hieroglyphicarum Inscriptionum Cretae)	Published 1996
Cypro-Minoan	DĀMOS (Oslo)	Partial; work in progress
Linear Elamite	Desset et al. (2022)	Recent partial decipherment publication

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Glossa Lab Implementation Status

The following scripts have **active corpus modules** implemented in `backend/glossa_lab/data/`:

Script	Module	Corpus size	Benchmark	Status
Proto-Sinaitic	`proto_sinaitic.py`	576 tokens, 22 signs	Tier 1e	☑ Active — 19/22 = 86.4% with anchors
Meroitic	`meroitic.py`	551 tokens, 19 signs	Tier 1f	☑ Active — graceful degradation test
Indus / Harappan	`indus_public_corpus.py`	14,213 tokens, 713 signs	Tier 5	☑ Active — primary research target

The following are implemented as **test fixtures** or **language model corpora**:

Script	Usage	Location
Ugaritic	Tier 1a/1b/1c cipher text	`tests/corpora/ugaritic.py`
Old Hebrew	Tier 1a/1b/1c target LM	`data/old_hebrew.py`
Phoenician	Tier 1c target LM	`data/phoenician.py`
Linear B	Tier 4 (Ventris)	`tests/corpora/fixtures/linear_b.txt`

Script	Usage	Location
Tamil/Dravidian	Tier 5 comparison LM	`data/dravidian.py`
Sumerian	Tier 3 logographic reference	`tests/corpora/fixtures/sumerian.txt`

Digital Corpus Availability — Detailed Notes

High-priority scripts with substantial digital corpora

Indus / Harappan Script

- ICIT corpus (Fuls 2023): ~4,410 inscriptions, 14,213 tokens — held by Dr. Fuls, TU Berlin
- Mahadevan concordance M77: ~3,700 inscriptions; partially OCR'd in `reports/`
- IVC Digital Database (BSOAS): available to academic subscribers
- Format: sign ID sequences (Fuls numbering or Mahadevan M-codes)
- Current Glossa Lab status: full pipeline active on ICIT corpus

Proto-Elamite

- CDLI (Cuneiform Digital Library Initiative): ~5,000 tablets at cdli.ucla.edu
- Accessible via API: `https://cdli.ucla.edu/search/`
- Format: cuneiform sign sequences in ATF transliteration
- Token estimate: ~200,000 sign tokens across all tablets
- Note: purely administrative (numerals + commodities), no phonetic reading possible
- Glossa Lab status: not yet implemented; high priority for structural analysis

Linear A

- SigLA (Signary of Linear A): sigla.classics.ox.ac.uk — ~1,500 inscriptions
- GORILA (Recueil des inscriptions en linéaire A): published 1985–2003
- Digital: DĀMOS (Oslo) partial; tylerlengyel.com Phase 1 data (used in experiments)
- Format: sign code sequences (AB01, AB02, etc.)
- Token estimate: ~7,500–10,000 sign tokens
- Glossa Lab status: statistical model in `tests/corpora/linear_a_corpus.py`

Meroitic

- REM (Répertoire d'épigraphie méroïtique): ~1,200 inscriptions; publication ongoing
- Digital: partial transcriptions at academia.edu; Rilly (2007) appendix
- Format: sign sequences (Griffith values: a, e, i, b, t, k, n, r, s, l, m, w, y, d, q, h, ne, se, te)
- Token estimate: ~15,000–20,000 sign tokens from attested corpus
- Glossa Lab status: active in `data/meroitic.py` (551-token sample corpus)

Rongorongo

- Fischer (1997) catalogue: 25 surviving objects, ~14,000 glyphs total
- Digital: Hochenburger database; Fischer 1997 ISBN 978-0-252-02349-9
- Format: glyph ID sequences (Fischer numbering)
- Token estimate: ~3,000–4,000 unique glyph types; 14,000 total tokens

- Note: corpus is very small per inscription; ~25 objects means ~25 sequences
- Glossa Lab status: not yet implemented; blocked by small per-inscription corpus

Cretan Hieroglyphic

- CHIC (Corpus Hieroglyphicarum Inscriptionum Cretae, Olivier & Godart 1996)
- ~300 inscribed objects; digitised from CHIC publication
- Token estimate: ~800–1,200 sign tokens
- Glossa Lab status: too small for statistical methods; pending

Cypro-Minoan

- DĀMOS (Database of Mycenaean at Oslo): partial
- Steele (2018) corpus: most comprehensive
- ~230 inscriptions; ~2,000 sign tokens
- Glossa Lab status: candidate for Tier 1 if Linear B phoneme overlap is used as anchor

Scripts needing corpus acquisition

Proto-Sinaitic (expanded)

- Current Glossa Lab corpus: 576 tokens (research-quality reconstruction)
- Full attested corpus: ~40 inscriptions, ~500 sign tokens
- Available: Sass (1988), Hamilton (2006), Darnell et al. (2005)
- Status: ADEQUATE — our 576-token corpus exceeds the attested material

Linear Elamite (newly partially deciphered)

- Desset et al. (2022) partial decipherment — Nature paper
- ~40 inscriptions; ~500 sign tokens
- Acquisition: the paper appendices and supplementary material
- Status: high priority — partial anchor points now available

Khitan (Large and Small)

- Several hundred texts; some catalogued in Chinese academic databases
- Format: character sequences with some known morphological markers
- Status: difficult acquisition; Chinese academic partnerships needed

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Corpora Planned for Future Implementation

Based on acquisition feasibility and research priority:

Script	Priority	Blocker	Expected Tier
Proto-Elamite	High	Format conversion from CDLI ATF	Tier 3 (structural only)
Linear Elamite	High	Acquire Desset 2022 supplement	Tier 1 (with partial anchors)
Rongorongo	Medium	Acquire Fischer 1997 digital	Tier 5 (Polynesian LM)
Cypro-Minoan	Medium	DĀMOS access	Tier 1 (Linear B anchors)

Script	Priority	Blocker	Expected Tier
Vinca symbols	Low	Disputed writing status	Structural only
Khitan Large	Low	Chinese database access	Structural only

Last updated: 2026-04-10. See `AGENTS.md` for experiment-addition procedures, `LEDGER.md` for research history, and `docs/FINETUNING_GUIDE.md` for AI model tuning. Cross-reference: `backend/glossa_lab/data/` for implemented corpus modules.